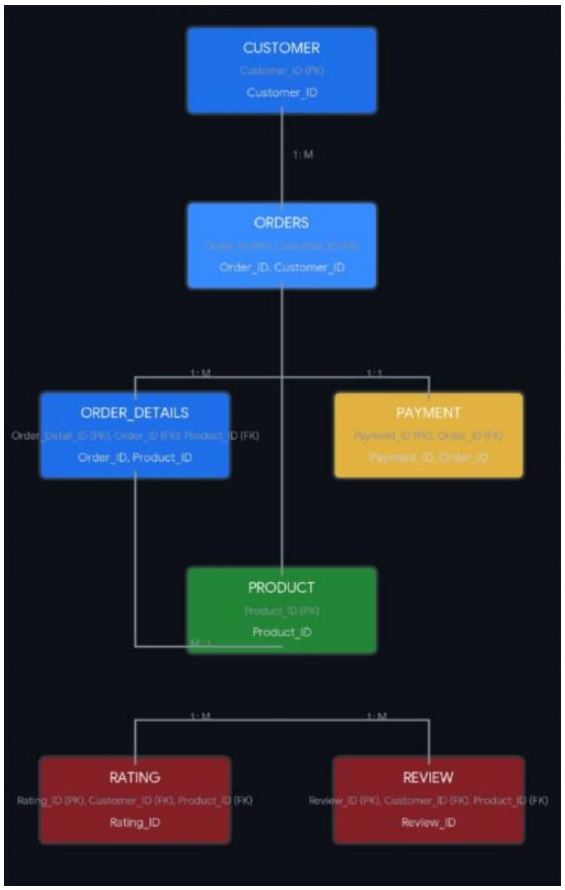
Zara Fashion E-Commerce Management System

DATABASE RELATIONSHIP ANALYSIS USING JOINS

CHAPTER-9

1.Combining Multiple Tables

Combining multiple tables using JOIN operations helps retrieve related data from different tables in the Zara Fashion E-Commerce Database based on common columns.



2.INNER JOIN

The INNER JOIN selects and returns only the rows that have matching values in both tables.



2.1 Customer and Orders

```
SQL> SELECT C.CUSTOMERID,
  2         C.AGE,
  3         C.GENDER,
  4         C.PHONE,
  5         O.ORDER_ID,
  6         O.ORDER_DATE,
  7         O.TOTAL_AMOUNT
  8  FROM CUSTOMER C
  9  INNER JOIN ORDERS O
 10  ON C.CUSTOMERID = O.CUSTOMER_ID
 11  ORDER BY O.ORDER_ID;

CUSTOMERID        AGE GENDER     PHONE             ORDER_ID ORDER_DAT
---------- ---------- ---------- --------------- ---------- ---------
TOTAL_AMOUNT
------------
       224         18 female     9840472788             101 01-OCT-26
        4999

       108         19 female     9884752129             102 02-OCT-26
        2999

       224         18 female     9840472788             103 03-OCT-26
        1999
```

2.2 Orders and payment

```
SQL> SELECT
  2      O.ORDER_ID,
  3      P.PAYMENT_ID,
  4      P.PAYMENT_MODE,
  5      P.PAYMENT_DATE,
  6      P.PAYMENT_AMOUNT,
  7      P.PAYMENT_STATUS
  8  FROM ORDERS O
  9  INNER JOIN PAYMENT P
 10      ON O.ORDER_ID = P.ORDER_ID;

  ORDER_ID PAYMENT_ID PAYMENT_MODE        PAYMENT_D PAYMENT_AMOUNT
---------- ---------- -------------------- --------- --------------
PAYMENT_STATUS
--------------------
       101        501 UPI                  29-SEP-26           1898
Successful

       102        502 Credit Card          29-SEP-26           1299
Successful

       103        503 Debit Card           29-SEP-26           1499
Successful


  ORDER_ID PAYMENT_ID PAYMENT_MODE        PAYMENT_D PAYMENT_AMOUNT
---------- ---------- -------------------- --------- --------------
PAYMENT_STATUS
```

2.3 Product and Order details

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      OD.ORDER_ID,
  5      OD.QUANTITY,
  6      OD.UNIT_PRICE,
  7      OD.SUBTOTAL
  8  FROM PRODUCT P
  9  INNER JOIN ORDER_DETAILS OD
 10      ON P.PRODUCT_ID = OD.PRODUCT_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
  ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ----------
       101
Floral Dress
       101          2       2499       4998

       102
Casual T-Shirt
       101          1        999        999

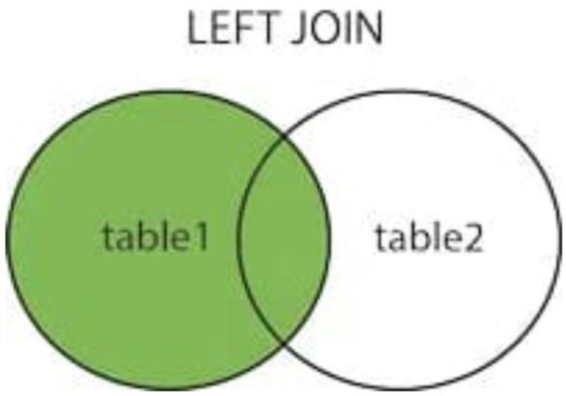
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
  ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ----------

       103
Slim Fit Jeans
       102          1       1999       1999

       104
Running Shoes
```

3.LEFT JOIN

LEFT JOIN displays all records from the left table and matching records from the right table; if there is no match, NULL is displayed.



3.1 Display all customers and their orders

```
SQL> SELECT
  2      c.CUSTOMERID,
  3      c.AGE,
  4      c.GENDER,
  5      o.ORDER_ID,
  6      o.ORDER_DATE,
  7      o.TOTAL_AMOUNT
  8  FROM CUSTOMER c
  9  LEFT JOIN ORDERS o
 10      ON c.CUSTOMERID = o.CUSTOMER_ID;

CUSTOMERID        AGE GENDER       ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- ---------- ---------- ---------- --------- ------------
       224         18 female            101 01-OCT-26         4999
       108         19 female            102 02-OCT-26         2999
       224         18 female            103 03-OCT-26         1999
       567         20 female            104 04-OCT-26         5999
       108         19 female            105 05-OCT-26         3499
```

4.RIGHT JOIN

```
SQL> SELECT
  2      c.CUSTOMERID,
  3      c.AGE,
  4      c.GENDER,
  5      o.ORDER_ID,
  6      o.ORDER_DATE,
  7      o.TOTAL_AMOUNT
  8  FROM CUSTOMER c
  9  RIGHT JOIN ORDERS o
 10      ON c.CUSTOMERID = o.CUSTOMER_ID;

CUSTOMERID        AGE GENDER       ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- ---------- ---------- ---------- --------- ------------
       224         18 female            101 01-OCT-26         4999
       108         19 female            102 02-OCT-26         2999
       224         18 female            103 03-OCT-26         1999
```

4.1 Display all orders and customer details

```
SQL> SELECT
  2      C.CUSTOMERID,
  3      O.CUSTOMER_ID,
  4      O.ORDER_ID,
  5      O.ORDER_DATE,
  6      P.PRODUCT_ID,
  7      P.PRODUCT_NAME,
  8      OD.QUANTITY,
  9      OD.UNIT_PRICE,
 10      OD.SUBTOTAL,
 11      PY.PAYMENT_MODE,
 12      PY.PAYMENT_AMOUNT,
 13      PY.PAYMENT_STATUS
 14  FROM CUSTOMER C
 15  INNER JOIN ORDERS O
 16      ON C.CUSTOMERID = O.CUSTOMER_ID
 17  INNER JOIN ORDER_DETAILS OD
 18      ON O.ORDER_ID = OD.ORDER_ID
 19  INNER JOIN PRODUCT P
 20      ON OD.PRODUCT_ID = P.PRODUCT_ID
 21  LEFT JOIN PAYMENT PY
 22      ON O.ORDER_ID = PY.ORDER_ID
 23  ORDER BY O.ORDER_DATE DESC;
```

```
CUSTOMERID CUSTOMER_ID   ORDER_ID ORDER_DAT PRODUCT_ID
---------- ----------- ---------- --------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL PAYMENT_MODE         PAYMENT_AMOUNT
---------- ---------- ---------- -------------------- --------------
PAYMENT_STATUS
--------------------
       108         108        105 05-OCT-26        104
Running Shoes
         1       2999       2999 Cash on Delivery                999
Failed


CUSTOMERID CUSTOMER_ID   ORDER_ID ORDER_DAT PRODUCT_ID
---------- ----------- ---------- --------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL PAYMENT_MODE         PAYMENT_AMOUNT
---------- ---------- ---------- -------------------- --------------
PAYMENT_STATUS
--------------------
       567         567        104 04-OCT-26        110
Hand Bag
         1       2499       2499 UPI                          2499
Successful
```

5. Complete Order details

```
SQL> SELECT
  2      O.ORDER_ID,
  3      O.CUSTOMER_ID,
  4      O.ORDER_DATE,
  5      O.TOTAL_AMOUNT,
  6      P.PRODUCT_ID,
  7      P.PRODUCT_NAME,
  8      OD.QUANTITY,
  9      OD.UNIT_PRICE,
 10      OD.SUBTOTAL,
 11      PY.PAYMENT_MODE,
 12      PY.PAYMENT_AMOUNT,
 13      PY.PAYMENT_STATUS
 14  FROM ORDERS O
 15  INNER JOIN ORDER_DETAILS OD
 16      ON O.ORDER_ID = OD.ORDER_ID
 17  INNER JOIN PRODUCT P
 18      ON OD.PRODUCT_ID = P.PRODUCT_ID
 19  LEFT JOIN PAYMENT PY
 20      ON O.ORDER_ID = PY.ORDER_ID
 21  ORDER BY O.ORDER_DATE DESC;

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT PRODUCT_ID
---------- ----------- --------- ------------ ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL PAYMENT_MODE         PAYMENT_AMOUNT
---------- ---------- ---------- -------------------- --------------
PAYMENT_STATUS
--------------------
       105         108 05-OCT-26         3499        104
Running Shoes
         1       2999       2999 Cash on Delivery                999
Failed


  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT PRODUCT_ID
---------- ----------- --------- ------------ ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
```

6.Customer purchase history

```
SQL> SELECT
  2      C.CUSTOMERID,
  3      C.AGE,
  4      C.GENDER,
  5      O.ORDER_ID,
  6      O.ORDER_DATE,
  7      P.PRODUCT_ID,
  8      P.PRODUCT_NAME,
  9      OD.QUANTITY,
 10      OD.UNIT_PRICE,
 11      OD.SUBTOTAL,
 12      O.TOTAL_AMOUNT,
 13      PY.PAYMENT_MODE,
 14      PY.PAYMENT_STATUS
 15  FROM CUSTOMER C
 16  INNER JOIN ORDERS O
 17      ON C.CUSTOMERID = O.CUSTOMER_ID
 18  INNER JOIN ORDER_DETAILS OD
 19      ON O.ORDER_ID = OD.ORDER_ID
 20  INNER JOIN PRODUCT P
 21      ON OD.PRODUCT_ID = P.PRODUCT_ID
 22  LEFT JOIN PAYMENT PY
 23      ON O.ORDER_ID = PY.ORDER_ID
 24  ORDER BY C.CUSTOMERID, O.ORDER_DATE DESC;

CUSTOMERID        AGE GENDER       ORDER_ID ORDER_DAT PRO
---------- ---------- ---------- ---------- --------- ---
PRODUCT_NAME
--------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL TOTAL_AMOUNT PAYMENT_MOD
---------- ---------- ---------- ------------ -----------
PAYMENT_STATUS
--------------------
       108         19 female            105 05-OCT-26
Running Shoes
```

7.Customer purchase history with payment

```
SQL> SELECT
  2      C.CUSTOMERID,
  3      O.ORDER_ID,
  4      O.ORDER_DATE,
  5      O.TOTAL_AMOUNT,
  6      PY.PAYMENT_MODE,
  7      PY.PAYMENT_AMOUNT,
  8      PY.PAYMENT_STATUS
  9  FROM CUSTOMER C
 10  INNER JOIN ORDERS O
 11      ON C.CUSTOMERID = O.CUSTOMER_ID
 12  LEFT JOIN PAYMENT PY
 13      ON O.ORDER_ID = PY.ORDER_ID
 14  ORDER BY C.CUSTOMERID, O.ORDER_DATE DESC;

CUSTOMERID   ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_AMOUNT
---------- ---------- --------- ------------ -------------------- --------------
PAYMENT_STATUS
--------------------
       108        105 05-OCT-26         3499 Cash on Delivery               999
Failed

       108        102 02-OCT-26         2999 Credit Card                   1299
Successful

       224        103 03-OCT-26         1999 Debit Card                    1499
Successful


CUSTOMERID   ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_AMOUNT
---------- ---------- --------- ------------ -------------------- --------------
PAYMENT_STATUS
--------------------
       224        101 01-OCT-26         4999 UPI                           1898
Successful
```

8.Multi-Table business report

```
SQL> SELECT
  2      C.CUSTOMERID,
  3      COUNT(DISTINCT O.ORDER_ID) AS TOTAL_ORDERS,
  4      SUM(OD.SUBTOTAL) AS TOTAL_PURCHASE,
  5      SUM(OD.QUANTITY) AS TOTAL_PRODUCTS
  6  FROM CUSTOMER C
  7  INNER JOIN ORDERS O
  8      ON C.CUSTOMERID = O.CUSTOMER_ID
  9  INNER JOIN ORDER_DETAILS OD
 10      ON O.ORDER_ID = OD.ORDER_ID
 11  GROUP BY C.CUSTOMERID
 12  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMERID TOTAL_ORDERS TOTAL_PURCHASE TOTAL_PRODUCTS
---------- ------------ -------------- --------------
       224            2           8796              4
       108            2           6297              3
       567            1           5998              2
```

9.Payment transaction report

```
SQL> SELECT
  2      C.CUSTOMERID,
  3      O.ORDER_ID,
  4      O.ORDER_DATE,
  5      O.TOTAL_AMOUNT,
  6      PY.PAYMENT_MODE,
  7      PY.PAYMENT_AMOUNT,
  8      PY.PAYMENT_STATUS
  9  FROM CUSTOMER C
 10  INNER JOIN ORDERS O
 11      ON C.CUSTOMERID = O.CUSTOMER_ID
 12  INNER JOIN PAYMENT PY
 13      ON O.ORDER_ID = PY.ORDER_ID
 14  ORDER BY O.ORDER_DATE DESC;

CUSTOMERID   ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE        PAYMENT_AMOUNT
---------- ---------- --------- ------------ ------------------ --------------
PAYMENT_STATUS
--------------------
       108        105 05-OCT-26         3499 Cash on Delivery                999
Failed

       567        104 04-OCT-26         5999 UPI                             2499
Successful

       224        103 03-OCT-26         1999 Debit Card                      1499
Successful


CUSTOMERID   ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE        PAYMENT_AMOUNT
---------- ---------- --------- ------------ ------------------ --------------
PAYMENT_STATUS
--------------------
       108        102 02-OCT-26         2999 Credit Card                     1299
Successful
```